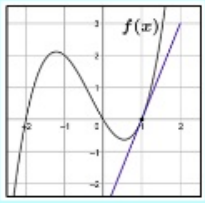
CHA-3.A.1 The derivative of a function can be interpreted as the instantaneous rate of change with respect to its independent variable.

CHA-3.A.2 The derivative can be used to express information about rates of change in applied contexts.

CHA-3.A.3 The unit for f '(x) is the unit for f divided by the unit for x.

4-1 The Derivative in Context

Get Ready...



The graph of $f(x)$ is at the right along with the line tangent to f through the point $(1, 0)$. It is known that $f'(1) = 3$. Interpret the meaning of the statement $f'(1) = 3$.

The rate of change of f at $x = 1$ is 3

We have been studying derivatives, including differentiation rules and the meaning of a derivative graphically. However, the problems we have been working with have been without context. Now, we will see examples of functions that have context, real world meaning. When functions have real world context, then the derivative will also take on real world context.

Important notes about contextual derivatives	#1: In real world context, we rarely use x and y for variables since they have no context. Instead, we use variables that correspond with the situation—often we use t for the horizontal axis (time). #2: In contextual problems, derivatives have units: $\frac{\text{units of } y}{\text{units of } x}$ #3: In real world context, a derivative is a **RATE**. (rate ~ slope) #4: Derivatives are still **INSTANTANEOUS**—they occur at an instant! (...at t = 3 minutes)

Derivatives are…	Which means …	And NOT …
rates	"The rate of …"	~~Totals, Amounts, or Averages~~
instantaneous	"at t = …"	~~Intervals of time~~
contextual with units	Context with units $\frac{\text{units of } y}{\text{units of } x}$	~~Units of original function~~

Example 1: The height of a sunflower, in inches, is measured over the course of 7 weeks. Use the table above to estimate $H'(2)$. Include units of measure.

t (weeks)	1	3	4	7
$H(t)$ (inches)	7	25	33	62

$$H'(2) \approx \frac{H(3)-H(1)}{3-1} = \frac{25-7}{2} = \frac{18}{2} = 9 \text{ in/wk}$$

Example 2: The number of cars that enter the parking lot of a grocery store t hours after it opens is changing at a rate modeled by the function $C'(t)$,where $C'(t)$ is measured in cars per hour. Use the table above to approximate $C''(6)$. Include units of measure.

t (hours)	0	2	5	7
$C'(t)$ (cars per hour)	28	65	92	54

$$C''(6) \approx \frac{C'(7)-C'(5)}{7-5} = \frac{54-92}{2} = \frac{-38}{2} = -19 \ \frac{\text{cars per hr}}{\text{hr}}$$

Interpreting a Derivative in Context	To interpret a derivative in context, be sure to include the following: 1. Context—don't just name the function…include what the function is modeling with words! 2. Units—Include the units for BOTH the answer and the x variable (often it is time) 3. Instantaneous time—must be clear: at $t = \cdots$ 4. Rate—include the word rate in your sentence.

Example 3: Interpret the meaning of your answer in context from **Example 1** above.

t (weeks)	1	3	4	7
$H(t)$ (inches)	7	25	33	62

The height of a sunflower, in inches, is measured over the course of 7 weeks. Use the table above to estimate $H'(2)$. Include units of measure.

$$H'(2) = \frac{H(3)-H(1)}{3-1} = \frac{25-7}{3-1} = \frac{18}{2} = 9 \text{ inches/week}$$

Interpreting a Derivative in Context

Interpret $f'(c)$ given $f(t)$

The **height of the sunflower** (*f(t)* in context) is **increasing** (increasing/decreasing) at a ***rate*** of **9 in/wk** (|*f'(c)*| with units) at $t =$ **2 weeks** (time with units).

Example 4: A factory produces delicious cans of corn for the typical corn enthusiast. One particular machine is notorious for packaging cans of corn at impressive speeds. The amount of cans this machine has packaged after t seconds is modeled by the differentiable function $C(t)$.

t (sec)	0	2	3	6
$C(t)$ (cans)	16	52	71	125

Approximate $C'(1)$. Using correct units, interpret the meaning of this value in context of the problem.

$$C'(1) \approx \frac{C(2)-C(0)}{2-0} = \frac{52-16}{2} = \frac{36}{2} = 18 \text{ cans/sec}$$

$C'(1)$ means the amount of cans packaged is increasing at a rate of 18 cans/sec at $t=1$ second.

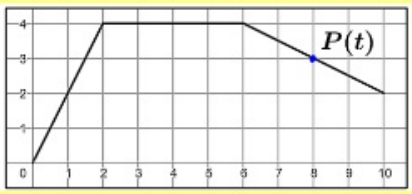
Example 5: (Calculator active) At the end of a football game, fans begin to exit the stadium. For $0 \le t \le 40$, the number of fans that have exited the stadium after t minutes can be modeled by the function $F(t) = \frac{50{,}000}{1+535e^{-0.29t}}$. Find $F'(8)$. Using correct units, interpret the meaning of this value in context of the problem.

$F'(8) = 265.590$ Fans/min

$\frac{\text{units for } f(t)}{\text{Units for } t}$

$F'(8)$ means the number of fans that have exited the stadium is increasing at a rate of 265.590 fans/min at $t=8$ minutes

Example 6: For $0 < t < 10$, the rate that people are getting on an escalator, in people per second, is modeled by the function P, where t is measured in seconds.



Use the graph of P to find $P'(8)$. Using correct units,interpret the meaning of this value in context of the problem.

$P'(8) = -\frac{1}{2}$ people per sec/sec

$P'(8)$ means the rate people are getting on an escalator is decreasing at a rate of $\frac{1}{2}$ people per sec/sec at $t = 8$ sec.